

Where Two-Stage Cover Song Retrieval Loses Its Covers: Candidate Recall, Hubness and Alignment Cost on Da-TACOS

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Abstract

Cover song identification (CSI) must recognise a composition across changes of key, tempo, instrumentation and structure. Structure-aware alignment handles these changes well but costs quadratic time per pair, so scalable systems shortlist K candidates with a learned embedding and align only those. Alignment cannot recover a cover that the shortlist excludes. We measure this constraint on all 13,000 queries of the Da-TACOS benchmark with a transparent HPCP system: a compact temporal convolutional encoder, then key-rotated subsequence dynamic time warping over the shortlist. Every setting was locked on held-out works before the benchmark was run. Because Hit@1 factorises exactly into shortlist coverage and reranker efficiency, we can assign every query’s failure to one stage. At the deployed $K = 30$, 47.8% of queries have no cover in the shortlist. Across four locked evaluations, coverage accounts for about 80% of the log-gain in a $3.6\times$ improvement of Hit@1. The gap to exhaustive 96-frame alignment narrows from $4.5\times$ to 10% in MAP, with 30 instead of 14,999 alignments per query. Enlarging the shortlist raises MAP from 0.122 to 0.212 at $K = 500$, but shifts the dominant failure from exclusion to misranking between $K = 100$ and $K = 200$. Hub correction matters more as K grows: without it, a tenth of the catalogue supplies 33.8% of the rank-1 false positives in covered queries at $K = 500$. Two registered interventions on the coarse stage give opposite results. Per-query adaptive K does not beat fixed K at equal cost. Fusing global and window embeddings raises MAP at every K (+0.017 at $K = 30$). Finally, in four paired comparisons a 20-query development protocol was inconclusive every time and pointed the wrong way three times. Our absolute accuracy is well below published systems. The contribution is the measurement of where a two-stage system fails, not a new model.

1 Introduction

A cover preserves a composition’s identity while changing almost everything else: key, tempo, instrumentation, timbre, and often structure. Classical CSI handles these changes explicitly. It compares chroma sequences after choosing a transposition and aligns them in time [Serrà et al., 2008, 2009]. Alignment is accurate but costs $O(NM)$ per pair, and a catalogue of n tracks needs n^2 pairs. On the Da-TACOS benchmark [Yesiler et al., 2019] that is 1.95×10^8 query–candidate pairs. Learned CSI replaces alignment with a single embedding per recording, compared by cosine similarity [Yesiler et al., 2020a, Du et al., 2021, Araz et al., 2024]. It is fast but blurs temporal structure.

The natural compromise is *coarse-to-fine* retrieval. An embedding shortlists K candidates per query, and structure-aware alignment reranks only those. ByteCover3 retrieves with approximate nearest neighbours and reranks with local features [Du et al., 2023]; LIVI reranks with late interaction over local embeddings [Affolter et al., 2026]; text retrieval has used cascades for over a decade [Wang et al., 2011]. The design has an obvious ceiling: a cover the coarse stage excludes cannot be recovered. What is not obvious is *how large* that ceiling is at realistic shortlist sizes, how much of a system’s improvement comes from raising it, and when it stops being the main limit. We found no CSI study that reports these quantities.

We measure them on the full Da-TACOS benchmark (13,000 queries, 15,000 candidates, 12 relevant items per query). The system is deliberately transparent: HPCP features, a 645,504-parameter temporal convolutional encoder, exact cosine search, key rotation by chroma-profile cosine, and slope-constrained subsequence DTW. Every hyperparameter was locked on held-out works before the benchmark was run. The system is far from state-of-the-art accuracy. What it offers is complete control: exact per-query ranks for every stage, four locked benchmark evaluations, a sweep of K from 5 to 500 that reproduces the locked run exactly at $K = 30$, and paired, work-level statistics.

Contributions.

1. **A per-query decomposition of two-stage CSI.** Hit@1 factorises exactly into shortlist coverage and reranker efficiency. We use this to attribute a $3.6\times$ Hit@1 improvement mostly to the coarse stage, and to locate the shortlist size ($K \approx 100\text{--}200$) where misranking overtakes exclusion as the main failure (Sections 6–8).

2. **A controlled study of what moves each stage:** training data and training length, window fusion in Stage 1, alignment resolution, key handling, and hubness correction, whose benefit grows with K . It includes a registered negative result for per-query adaptive K (Sections 7–8).
3. **An accuracy–computation analysis** that separates what is measured from what was never run (Section 9).
4. **A paired demonstration that a 20-query development protocol cannot substitute for benchmark evaluation** in this setting, with an explanation in terms of protocol structure (Section 10).

We do not claim a new architecture, a new hubness correction, or state-of-the-art accuracy. Two-stage retrieval, local features and hubness correction are all prior art (Section 2).

2 Related Work

Classical CSI.

- Harmonic pitch class profiles (HPCP) [Gómez, 2006] are the pitch-class features Da-TACOS distributes.
- Serrà et al. choose an optimal transposition index and align binarised chroma locally [Serrà et al., 2008]; Qmax quantifies cross-recurrence [Serrà et al., 2009].
- Our Stage 2 uses the same ingredients in simpler form: profile-based key choice and whole-query subsequence DTW [Sakoe and Chiba, 1978, Müller, 2015].
- Scalability has been a concern since at least Bertin-Mahieux and Ellis [2012].

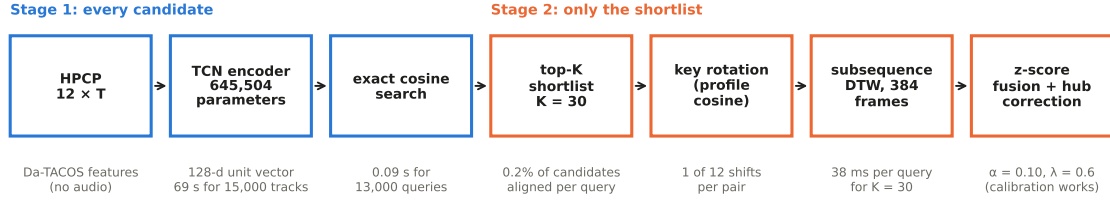
Learned version identification.

- MOVE and Re-MOVE learn embeddings from pitch-class input [Yesiler et al., 2020a,b].
- ByteCover and ByteCover2 train CQT encoders with multiple losses [Du et al., 2021, 2022]. CoverHunter adds attention and coarse-to-fine alignment during training [Liu et al., 2023].
- Discogs-VI provides 1.9 million versions and an open model [Araz et al., 2024]. CLEWS learns segment-level representations [Serrà et al., 2025].
- Our encoder combines a generic temporal convolutional network [Bai et al., 2018] with a supervised contrastive loss [Khosla et al., 2020]. Neither component is new.

Two-stage and local retrieval. ByteCover3 uses local features and a two-stage pipeline of ANN retrieval and re-ranking [Du et al., 2023]. LIVI shortlists with track embeddings and re-ranks with MaxSim over local embeddings [Affolter et al., 2026], following late interaction in text retrieval [Khattab and Zaharia, 2020]. Multi-stage IR studies the cost–effectiveness trade-off of cascades [Wang et al., 2011], predicts per-query candidate depth [Culpepper et al., 2016], and reports that rerankers degrade as they score more candidates [Jacob et al., 2024]. Our work sits between these lines. We do not propose a stage; we measure what the interface between the stages costs in CSI.

Hubness. Some items are near neighbours of disproportionately many queries. This is known in audio similarity [Aucouturier and Pachet, 2008] and in high-dimensional spaces generally [Radovanović et al., 2010, Schnitzer et al., 2012]. Corrections that subtract a candidate’s typical similarity include CSLS [Conneau et al., 2018] and querybank normalisation [Bogolin et al., 2022]. In CSI, hubness has been reduced with mutual proximity [Li and Chen, 2018], at feature summarisation [Hu and Chen, 2019], and by normalising alignment similarity with a hubness score [Seo, 2022]. Our correction is of this family.

Benchmarks. Da-TACOS [Yesiler et al., 2019] distributes features, not audio, and defines the 15,000-track benchmark used here. Recent work moves to much larger and noisier collections: Discogs-VI [Araz et al., 2024]; DiVers, with more than a million versions including in-the-wild YouTube audio [Hachmeier et al., 2026]; studies of robustness to real-world covers [Hachmeier and Jäschke, 2025]; and joint track and version identification benchmarks [Araz et al., 2026]. MIREX CSI results use private collections and are not comparable with Da-TACOS numbers, so we do not quote them.



values: reports/results/benchmark_long384.json (run 4)

Figure 1: The two-stage system with run-4 settings and measured costs. Stage 1 scores every candidate; Stage 2 aligns only the K -item shortlist (0.2% of candidates at $K = 30$).

3 Problem Formulation

Let $\mathcal{C}$ be the candidate set ($|\mathcal{C}| = 15,000$). For query q , let $\mathcal{R}_q \subset \mathcal{C} \setminus \{q\}$ be its relevant set ($|\mathcal{R}_q| = 12$). Stage 1 induces a ranking $\pi_q^{(1)}$ of $\mathcal{C} \setminus \{q\}$. Let

$$r_q = \min_{c \in \mathcal{R}_q} \pi_q^{(1)}(c)$$

be the Stage-1 rank of the first relevant item.

A K -candidate reranker permutes the first K items of $\pi_q^{(1)}$ and leaves the rest unchanged.

Coverage and efficiency. Query q is *covered* at K iff $r_q \leq K$. Define

$$C(K) = \Pr_q[r_q \leq K] \quad (\text{coverage}),$$

$$\eta(K) = \Pr_q[\text{rank-1 item is relevant} \mid r_q \leq K] \quad (\text{reranker efficiency}),$$

$$\text{Hit@1}(K) = C(K) \eta(K) \quad (\text{exact}).$$

The identity holds because an uncovered query cannot have a relevant item at rank 1. Shortlist recall is $\text{Rec}(K) = \mathbb{E}_q[|\{c \in \mathcal{R}_q : \pi_q^{(1)}(c) \leq K\}|/|\mathcal{R}_q|]$. Coverage asks for at least one cover in the shortlist; recall asks for all 12.

Failure classes. Every query at a given K falls in exactly one class:

- **A:** not covered;
- **B:** covered, but a non-cover is at rank 1;
- **R:** a cover is at rank 1.

Within B we mark $\mathbf{B}_{\text{hub}}$ when the rank-1 item’s hub reference (Section 4) is above the 95th percentile of the catalogue, a threshold fixed before the analysis.

4 Method

Figure 1 shows the system; Table 1 lists every setting, read from configuration and result files.

Features. Each recording is a Da-TACOS HPCP sequence $X \in \mathbb{R}_{\geq 0}^{12 \times T}$. For alignment it is frame-normalised, area-resampled to $N \in \{96, 384\}$ frames, and frame-normalised again. The encoder view is resampled to 256 frames.

Stage 1: global embedding. A non-causal temporal convolutional network [Bai et al., 2018] maps X to a unit vector $e(X) \in \mathbb{R}^{128}$. It has six residual blocks with kernel 3 and dilations $1, \dots, 32$ (receptive field 253 of 256 frames), then mean and max pooling over time and an MLP head.

The encoder is trained with a supervised contrastive loss [Khosla et al., 2020] (temperature 0.1) on batches of 32 works $\times$ 2 recordings. Augmentation is random time crops (60–100%) and random cyclic pitch rotation. Optimisation is AdamW with cosine learning-rate decay; the checkpoint with the best validation MAP is kept.

Stage 1 scores $g(q, c) = e(X_q)^\top e(X_c)$ exactly over all candidates and keeps the top K .

Key handling. Let $\bar{p}(X)$ be the ℓ_2 -normalised time-average of X , and ρ_k cyclic rotation of the pitch axis by k semitones. We choose

$$k^*(q, c) = \arg \max_{k \in \{0, \dots, 11\}} \bar{p}(X_q)^\top \rho_k \bar{p}(X_c)$$

in the spirit of the optimal transposition index [Serrà et al., 2008], and align X_q with $\rho_{k^*} X_c$.

Subsequence DTW. With frame-wise cosine cost $C_{ij} = 1 - \cos(x_i^q, (\rho_{k^*} x^c)_j)$, the accumulated cost is

$$D_{i,j} = \min_{(a,b) \in \Sigma} D_{i-a, j-b} + w_{ab} C_{ij}, \quad \Sigma = \{(1, 1), (2, 1), (1, 2)\}, \quad w = (1, 2, 1),$$

with $D_{0,j} = C_{0,j}$ for all j , so the query may start anywhere in the candidate. The path ends at $\arg \min_j D_{N-1,j}$.

The step set bounds the path slope between $1/2$ and 2 , which rules out degenerate paths that match the whole query to one candidate frame. With these weights every admissible path has weighted length N , since each query frame is paid for once. Dividing the accumulated cost by the path weight therefore gives a normalised cost $\bar{D}$ whose minimiser is the same path.

The alignment score is $a(q, c) = 1 - \bar{D}(q, c)$.

Fusion. Within each shortlist, the scores are z-normalised and blended:

$$h(q, c) = \alpha z_q(g) + (1 - \alpha) z_q(a).$$

Candidates below the shortlist keep their Stage-1 order.

Hubness correction. A candidate that aligns well with everything gets a penalty:

$$a'(q, c) = a(q, c) - \lambda \bar{a}(c), \quad \bar{a}(c) = \frac{1}{|P|} \sum_{p \in P} a(p, c).$$

Here P is a fixed set of 200 probe queries from the *training* split, so no test query is involved. This is the candidate-side term of CSLS [Conneau et al., 2018]; the query-side term is constant within a query’s ranking and is omitted. Its cost is $200 \times 15,000$ extra alignments, done once.

Locked settings. α is chosen by grid search in steps of 0.05, λ from a grid, and the alignment resolution N is fixed, all on ten *calibration* works. Checkpoints are chosen on 150 validation works.

5 Experimental Protocol

5.1 Data and roles

Da-TACOS [Yesiler et al., 2019] provides HPCP features, not audio. Features and metadata are licensed CC BY-NC-SA 4.0, and nothing derived is redistributed here. Table 2 gives the work-disjoint roles:

- *training*: 4,780 works, or the first 1,500 of them for the smallest encoder;
- *validation*: 150 works, used for checkpoint selection;
- *calibration*: 10 works, used for α , λ and N only;
- *development*: 20 query works, 40 distractor works;
- *benchmark*: the separate Da-TACOS benchmark subset.

Table 1: System configuration, read from configs/base.yaml and the locked block of each benchmark result file

component	setting
input	Da-TACOS HPCP, 12 pitch classes; encoder view area-pooled to 256 frames
encoder	TCN, 6 residual blocks, kernel 3, dilations 1-32, 128 channels, receptive field 253 frames
pooling and head	mean and max over time, MLP to 128-d, L2-normalised; 645,504 parameters
training	supervised contrastive (temperature 0.1), 32 works x 2 recordings per batch, AdamW lr 0.001, cosine schedule, crop 0.6-1.0, random pitch rotation, seed 20260817
Stage 1	exact cosine search over all candidates, self excluded, top K = 30
key handling	one of 12 rotations, chosen by cosine of time-averaged chroma profiles
alignment	cosine cost, subsequence DTW, steps [[1, 1], [2, 1], [1, 2]], weights [1.0, 2.0, 1.0], cost / path weight
fusion	$\alpha z(\text{global}) + (1 - \alpha) z(\text{alignment})$ within the shortlist
hub correction	alignment score minus λ x mean score against 200 training-split probe queries
Run 1 locked	checkpoint sweep_base, alignment frames 96, α 0.05, λ -
Run 2 locked	checkpoint encoder_full_train, alignment frames 96, α 0.15, λ 0.5
Run 3 locked	checkpoint encoder_full_train, alignment frames 384, α 0.15, λ 0.6
Run 4 locked	checkpoint encoder_full_long, alignment frames 384, α 0.1, λ 0.6

Table 2: Data and evaluation protocol (all splits work-disjoint)

split	works	recordings	used for
Cover Analysis · train	4,780	9,560	encoder training
Cover Analysis · validation	150	300	checkpoint selection
Cover Analysis · calibration	10	20	α , λ , resolution; nothing else
Cover Analysis · dev queries + distractors	60	120	development protocol (20×120)
Benchmark	3,000	15,000	test only: 13,000 queries, 2,000 noise tracks, 12 relevant per query

5.2 Benchmark

All 13,000 clique recordings ($1,000 \text{ works} \times 13$) are queries against all 15,000 tracks, which include 2,000 noise tracks, with self-matches excluded. Each query therefore has 12 relevant items among 14,999 candidates. The Cover Analysis and benchmark subsets share no work and no recording (0 common WIDs, 0 common PIDs; DATASET_CARD.md), so training cannot leak into the benchmark. The benchmark was used only for final evaluation. No setting was chosen on it, and no locked run was repeated after its result was seen.

5.3 Development protocol

20 queries each rank a pool of 119 candidates containing exactly one relevant item, so AP equals reciprocal rank. The protocol was built as a fast plumbing check. Section 10 analyses what it can and cannot measure.

5.4 Metrics and statistics

We report MAP, MRR, Hit@ k , first-relevant rank, coverage $C(K)$ and shortlist recall.

Intervals are 95% paired bootstrap intervals with 10,000 resamples. We resample *query works* (1,000 cliques), not queries, because the 13 recordings of a work share relevance sets and are not independent [Efron and Tibshirani, 1993, Field and Welsh, 2007]. We call a difference supported only when its interval excludes zero. Intervals are per comparison and not adjusted for multiplicity; most supported benchmark differences exclude zero by a wide margin. Development intervals resample the 20 query works and are correspondingly wide.

5.5 Evidence tiers

Each result is labelled with how it was obtained:

- FROZEN: the four locked benchmark runs, with committed per-query files;
- DEV: the development protocol;

Table 3: Frozen benchmark systems: four separate evaluations (13,000 queries $\times$ 15,000 candidates)

run	system	MAP	MRR	Hit@1	Hit@10	med. rank	DTW pairs/query	ms/query
Run 1	global embedding	0.010	0.052	0.025	0.096	185	0	0.01
Run 1	hybrid $K=30$ $\alpha=0.05$	0.018	0.124	0.102	0.154	185	30	5.31
Run 1	classical alignment	0.084	0.282	0.244	0.352	77	14,999	727.56
Run 2	global embedding	0.034	0.130	0.072	0.243	59	0	0.01
Run 2	hybrid $K=30$ $\alpha=0.15$	0.058	0.247	0.205	0.318	59	30	5.34
Run 2	classical alignment	0.084	0.282	0.244	0.352	77	14,999	727.56
Run 2	hybrid hubcorr $K=30$ $\alpha=0.15$	0.062	0.271	0.234	0.330	59	30	5.34
Run 2	classical alignment hubcorr	0.136	0.397	0.350	0.481	14	14,999	727.56
Run 3	global embedding	0.034	0.130	0.072	0.243	59	0	0.01
Run 3	hybrid $K=30$ $\alpha=0.15$	0.062	0.271	0.236	0.326	59	30	38.03
Run 3	hybrid hubcorr $K=30$ $\alpha=0.15$	0.068	0.301	0.272	0.343	59	30	38.03
Run 4	global embedding	0.076	0.232	0.155	0.380	26	0	0.01
Run 4	hybrid $K=30$ $\alpha=0.10$	0.113	0.373	0.330	0.443	23	30	38.10
Run 4	hybrid hubcorr $K=30$ $\alpha=0.10$	0.122	0.402	0.366	0.462	21.5	30	38.10

- ANALYSIS: computed only from committed per-query files;
- LOCAL-FULL: later runs over all 13,000 queries on the author’s laptop, validated before use. At $K = 30$ they reproduce run 4 (match: exact; $|\Delta| = 0$ for all four systems’ MAP and for shortlist recall);
- POST-HOC: an analysis defined after its inputs were seen.

Runs A2 and E1 were registered, with success criteria, before their results existed.

6 Main Results

6.1 Four locked evaluations

Table 3 and Figure 2 report the four frozen runs.

- **Run 1** (1,500 training works, 60 epochs, 96-frame alignment): Stage 1 reaches 0.010 MAP and the hybrid 0.018. Exhaustive alignment reaches 0.084, $4.5\times$ better than the hybrid.
- **Run 2** trains the encoder on all 4,780 works: Stage 1 reaches 0.034, the hybrid 0.058, and the hub-corrected hybrid 0.062. Hub correction lifts exhaustive alignment to 0.136.
- **Run 3** reranks at 384 frames: 0.062; 0.068 with hub correction.
- **Run 4** trains for 150 epochs: Stage 1 reaches 0.076, the hybrid 0.113, and the hub-corrected hybrid 0.122. Reranking adds +0.0368 [+0.0340, +0.0395] MAP over Stage 1.

The best observed exhaustive system (0.136) remains ahead of the best hybrid (0.122). The two differ in resolution, though: 96 frames against 384. Exhaustive alignment at 384 frames was never run.

6.2 Coverage is the ceiling, and most gains came from raising it

Figure 3a shows $C(K)$ for the three encoders. At $K = 30$, 81.3% of queries had no cover in the shortlist with the first encoder, and 47.8% [46.2, 49.5] with the last. Shortlist recall rose from 0.021 to 0.136.

Across the four runs, coverage at $K = 30$ rose from 0.187 to 0.522 and efficiency from 0.546 to 0.701. Hit@1 rose $3.6\times$, from 0.102 to 0.366. Because $\log \text{Hit@1} = \log C + \log \eta$, about 80% of the log-gain is attributable to coverage (Figure 3b). Table 4 decomposes every query at $K = 30$. In run 4, class A (47.8%) is three times the size of class B (15.6%).

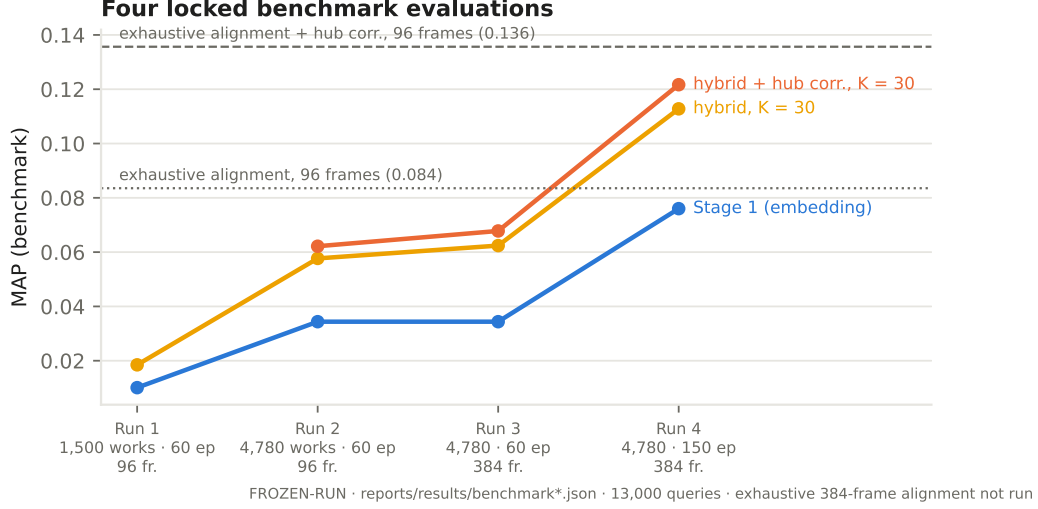


Figure 2: Benchmark MAP across the four locked evaluations. Each run changes one or two factors, listed under it. Horizontal lines: exhaustive alignment at 96 frames, with and without hub correction.

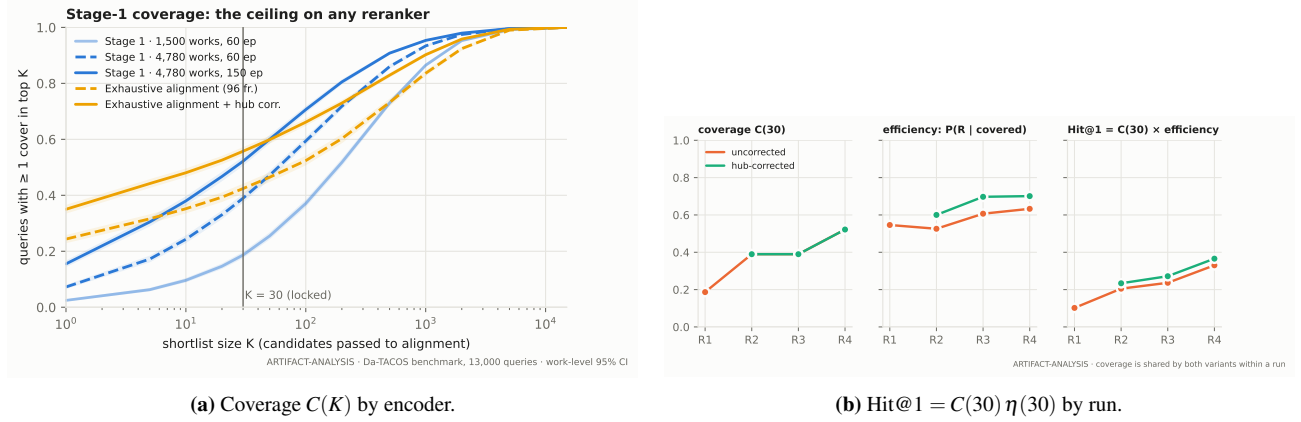


Figure 3: The coverage ceiling (ANALYSIS of the frozen runs).

6.3 Shortlist size: accuracy keeps rising, and the bottleneck moves

The frozen runs fix $K = 30$. Run A1 aligns the top 500 candidates of every query once and derives every $K \leq 500$ from that pass (LOCAL-FULL). Figure 4 and Table 5 give the results.

- **Accuracy.** Hub-corrected MAP rises from 0.085 at $K = 5$ to 0.122 at $K = 30$, 0.160 at $K = 100$ and 0.212 at $K = 500$. Hit@1 rises from 25.9% to 51.5%. Quality has not saturated at $K = 500$.
- **The bottleneck moves** (Figure 5, Table 6). Class A shrinks from 69.5% of queries to 9.2%, while class B grows from 4.6% to 39.3%. They are close at $K = 100$ (29.4% vs. 26.9%), and B dominates from $K = 200$ (33.1% vs. 19.5%).
- **Efficiency falls.** Reranker efficiency drops monotonically from 85.1% to 56.7%. Each additional candidate is one more chance to recover a cover and one more chance for a non-cover to outrank it.

Text rerankers show the same diminishing returns with depth [Jacob et al., 2024]; here it is measured for alignment reranking in CSI.

The claim that coverage is the dominant limitation therefore holds only for $K \lesssim 100$ in this system.

Table 4: Failure classes at K = 30, % of 13,000 queries (work-level 95% CI)

run	reranker	A: no cover in top 30	B: demoted	B: not promoted	R: rank 1	R given covered
Run 1	uncorrected	81.3 [80.2, 82.5]	3.0 [2.6, 3.4]	5.5 [5.0, 6.0]	10.2 [9.3, 11.1]	54.6 [51.8, 57.3]
Run 2	uncorrected	61.0 [59.5, 62.6]	6.7 [6.2, 7.2]	11.8 [11.1, 12.4]	20.5 [19.2, 21.9]	52.6 [50.5, 54.7]
Run 2	hub-corrected	61.0 [59.5, 62.6]	5.3 [4.9, 5.8]	10.2 [9.6, 10.9]	23.4 [22.0, 24.9]	60.0 [57.9, 62.1]
Run 3	uncorrected	61.0 [59.5, 62.6]	5.6 [5.2, 6.1]	9.7 [9.1, 10.3]	23.6 [22.2, 25.1]	60.7 [58.6, 62.8]
Run 3	hub-corrected	61.0 [59.5, 62.6]	4.1 [3.6, 4.5]	7.7 [7.2, 8.3]	27.2 [25.7, 28.7]	69.7 [67.8, 71.6]
Run 4	uncorrected	47.8 [46.2, 49.5]	8.3 [7.7, 8.9]	10.9 [10.3, 11.5]	33.0 [31.3, 34.7]	63.3 [61.4, 65.2]
Run 4	hub-corrected	47.8 [46.2, 49.5]	6.4 [5.9, 7.0]	9.2 [8.6, 9.8]	36.6 [34.9, 38.3]	70.1 [68.3, 71.8]

Table 5: Shortlist size K: coverage, recall and hub-corrected hybrid quality (LOCAL-FULL)

K	coverage	shortlist recall	MAP	Hit@1	Hit@10	rerank ms/query
5	0.305	0.048	0.0847	0.259	0.380	4
10	0.380	0.076	0.0949	0.299	0.380	7
20	0.467	0.111	0.1105	0.341	0.431	14
30	0.522	0.136	0.1217	0.366	0.462	22
50	0.600	0.173	0.1374	0.398	0.504	36
100	0.706	0.234	0.1599	0.438	0.553	72
200	0.805	0.312	0.1824	0.474	0.591	145
500	0.908	0.443	0.2116	0.515	0.634	362

Table 6: Failure classes by K, hub-corrected hybrid, % of queries (ANALYSIS of LOCAL-FULL)

K	A: no cover in top K	B: hub at rank 1	B: other	R: cover at rank 1	R given covered	B hub, uncorrected
5	69.5 [67.9, 71.1]	0.6 [0.4, 0.7]	4.0 [3.6, 4.4]	25.9 [24.4, 27.5]	85.1 [83.6, 86.5]	1.2
10	62.0 [60.4, 63.7]	0.9 [0.6, 1.1]	7.2 [6.6, 7.8]	29.9 [28.3, 31.6]	78.8 [77.1, 80.5]	2.0
20	53.3 [51.6, 55.0]	1.3 [1.0, 1.6]	11.4 [10.7, 12.1]	34.1 [32.3, 35.7]	72.9 [71.1, 74.6]	3.1
30	47.8 [46.2, 49.5]	1.5 [1.2, 1.8]	14.1 [13.3, 14.9]	36.6 [34.9, 38.3]	70.1 [68.3, 71.8]	3.9
50	40.0 [38.5, 41.5]	1.8 [1.5, 2.1]	18.5 [17.5, 19.4]	39.8 [38.1, 41.5]	66.3 [64.5, 68.0]	5.0
100	29.4 [28.1, 30.6]	2.2 [1.8, 2.5]	24.7 [23.7, 25.7]	43.8 [42.1, 45.4]	61.9 [60.2, 63.6]	6.8
200	19.5 [18.5, 20.4]	2.5 [2.1, 2.9]	30.7 [29.5, 31.9]	47.4 [45.7, 49.0]	58.8 [57.1, 60.5]	8.5
500	9.2 [8.6, 9.8]	2.8 [2.4, 3.2]	36.5 [35.1, 37.9]	51.5 [50.0, 53.1]	56.7 [55.1, 58.4]	11.2

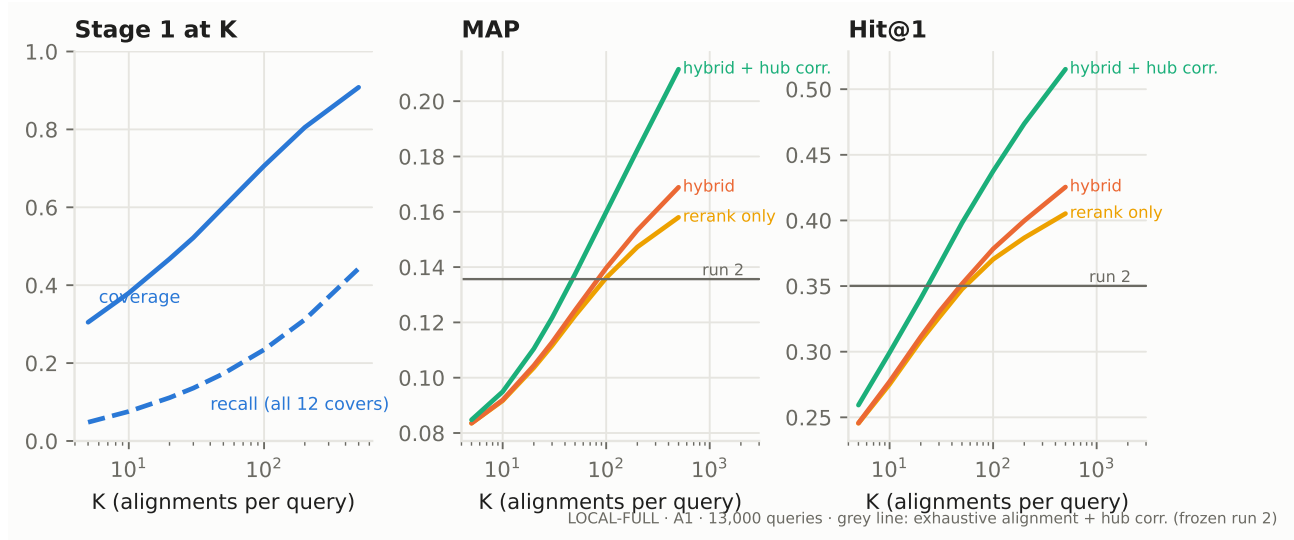


Figure 4: Shortlist size sweep (A1, LOCAL-FULL). Left: coverage and recall. Middle and right: MAP and Hit@1 of the hybrid, with and without hub correction. Grey: exhaustive corrected alignment at 96 frames (run 2).

7 Ablation Studies

Table 7 gives paired benchmark ablations from the frozen runs; Table 8 gives the alignment ablation at $K = 30$ (H1).

Training data and length. Training on 4,780 instead of 1,500 works raises validation MAP from 0.181 to 0.319. Training for 150 instead of 60 epochs raises it to 0.431 (best epoch 138). The 150-epoch curve rose between epochs 91–120 and 121–150 (means 0.409 and 0.421), then flattened: the fitted slope over the last 30 epochs is -0.00013 per epoch, as the cosine schedule decays the learning rate to zero. The data establish neither convergence nor continued improvement. In Stage-1 coverage at $K = 30$ (F1), the 60-epoch encoder is -13.2 points below the 150-epoch one and the 1,500-work encoder -33.5 points below. Training scale is the largest single effect we measured on the coarse stage.

Alignment resolution.

- DEV: 384 frames raise classical MAP from 0.248 to 0.384 and hybrid MAP from 0.262 to 0.362.
- Benchmark at $K = 30$ (run 3 vs. run 2): the effect is smaller but supported, +0.0047 [+0.0042, +0.0053] AP (Table 7).
- H1, rerank only at $K = 30$: 96 frames cost -0.0095 [-0.0105, -0.0084] relative to 384.

Temporal area-pooling to 96 frames averages away harmonic changes that alignment could otherwise match. The development result for *exhaustive* alignment at 384 frames has no benchmark counterpart.

Key handling (H1). Removing the key rotation costs -0.0386 [-0.0422, -0.0350] AP and -12.6 points of Hit@1. Trying all 12 rotations instead of the profile choice adds only +0.0015 [+0.0012, +0.0019] at twelve times the alignment cost.

Hubness correction. Hub correction improves every system (Table 7).

- It raises exhaustive alignment from 0.084 to 0.136.
- At $K = 30$ it converts 24.9% [23.0, 26.8] of class-B queries into successes, while breaking 1.2% of all queries.
- Its MAP gain grows with K : 0.001 at $K = 5$, 0.009 at $K = 30$, 0.043 at $K = 500$.

Section 8 explains why.

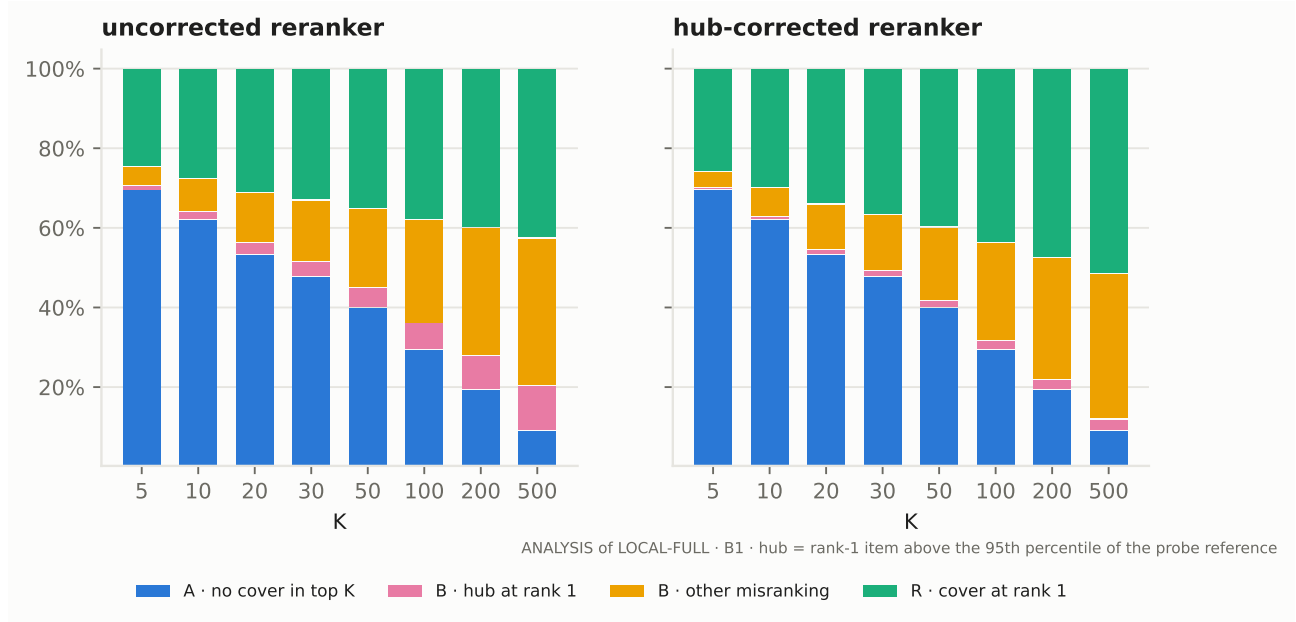


Figure 5: Failure classes by K without (left) and with (right) hub correction (B1). Blue: no cover in the shortlist. Pink and orange: covered but misranked. Green: success.

Changing Stage 1 without retraining (F1, A2).

- Test-time key rotation of the query adds +1.4 [+0.8, +2.0] points of coverage at $K = 30$.
- Maximum similarity over three overlapping 60%-length windows *loses* -1.5 points on its own.
- Averaging that window score with the global cosine adds +3.8 [+3.2, +4.5] points. The window count, lengths and the 50/50 weight were fixed in advance.

A registered end-to-end test of the fused Stage 1 (A2; criterion: paired Δ MAP above zero at $K = 30$ and $K = 100$) is met. MAP rises at every K we measured: +0.017 [+0.015, +0.019] at $K = 30$ and +0.014 [+0.013, +0.016] at $K = 100$. Hit@1 rises +3.0 points at $K = 30$. The gain is largest at small K , where coverage is the bottleneck. With fusion, $K = 20$ matches the MAP of the global Stage 1 at $K = 30$ (Figure 7, Table 9). Late interaction is not new [Khattab and Zaharia, 2020, Affolter et al., 2026]; the result shows that its coverage gain survives alignment reranking.

Allocating K per query (D1, E1). Label-free Stage-1 score statistics predict whether a query will be uncovered at $K = 30$, with cross-fitted AUROC 0.76 [0.75, 0.77]. A registered three-tier policy that sets K from this prediction, at equal mean cost, meets its criterion in 1 of 10 settings. Wherever it raised MAP it lowered Hit@1, so it is a trade-off, not an improvement (Appendix, Table 13). Adaptive candidate depth is established in IR [Culpepper et al., 2016]; it did not transfer here.

8 Failure Analysis

Classes at scale. Section 6 and Table 6 give the exact decomposition over all 13,000 queries. The hub subclass B_{hub} is small with correction at every K , but it grows without correction (last column of Table 6).

Why hubs appear: static candidates.

- **Mechanism.** A recording whose chroma barely changes over time has a nearly constant cost row against any query. Subsequence DTW then finds a cheap path through it regardless of the query’s content.

Table 7: Paired ablations on the benchmark ($\Delta = b - a$, work-level 95% CI)

comparison	setting	Δ AP	Δ Hit@1
resolution 96 to 384	same encoder (4,780 works, 60 ep), same alpha 0.15, same shortlist; 96 vs 384 frames, no hub correction	+0.0047 [+0.0042, +0.0053]	+0.031 [+0.027, +0.036]
resolution 96 to 384 hub	as above, both hub-corrected (lambda 0.5 at 96, 0.6 at 384, each from calibration)	+0.0056 [+0.0050, +0.0063]	+0.038 [+0.034, +0.042]
hub correction run2	same shortlist; alignment score minus lambda x probe reference	+0.0045 [+0.0040, +0.0050]	+0.029 [+0.025, +0.033]
hub correction run3	same shortlist; alignment score minus lambda x probe reference	+0.0054 [+0.0049, +0.0059]	+0.035 [+0.031, +0.039]
hub correction run4	same shortlist; alignment score minus lambda x probe reference	+0.0089 [+0.0081, +0.0097]	+0.036 [+0.031, +0.040]
hub correction classical	exhaustive alignment, 96 frames	+0.0521 [+0.0485, +0.0558]	+0.107 [+0.098, +0.115]
rerank vs stage1 run1	hybrid K = 30 against its own Stage 1	+0.0084 [+0.0075, +0.0094]	+0.077 [+0.070, +0.085]
rerank vs stage1 run2	hybrid K = 30 against its own Stage 1	+0.0233 [+0.0212, +0.0255]	+0.133 [+0.123, +0.142]
rerank vs stage1 run3	hybrid K = 30 against its own Stage 1	+0.0280 [+0.0256, +0.0305]	+0.164 [+0.153, +0.175]
rerank vs stage1 run4	hybrid K = 30 against its own Stage 1	+0.0368 [+0.0340, +0.0395]	+0.175 [+0.165, +0.186]
encoder data 1500 to 4780	Stage 1 only: 1,500 vs 4,780 training works, both 60 epochs	+0.0243 [+0.0215, +0.0272]	+0.048 [+0.041, +0.055]
encoder epochs 60 to 150	Stage 1 only: 60 vs 150 epochs, 4,780 works	+0.0416 [+0.0374, +0.0459]	+0.083 [+0.073, +0.092]
hybrid vs best classical	best hybrid (run 4) vs corrected exhaustive alignment (run 2): different runs, same queries	+0.0140 [+0.0075, +0.0204]	-0.016 [-0.027, -0.003]

Table 8: Alignment ablation at K = 30, rerank only (LOCAL-FULL); Δ vs the locked profile-cosine rotation at 384 frames

key handling	frames	MAP	Hit@1	Δ AP	Δ Hit@1
no key rotation	96	0.0713	0.189	-0.0402 [-0.0439, -0.0366]	-0.137 [-0.146, -0.128]
profile-cosine rotation (locked)	96	0.1021	0.288	-0.0095 [-0.0105, -0.0084]	-0.038 [-0.043, -0.034]
exhaustive 12-rotation DTW	96	0.1030	0.290	-0.0086 [-0.0097, -0.0076]	-0.036 [-0.041, -0.031]
no key rotation	384	0.0730	0.200	-0.0386 [-0.0422, -0.0350]	-0.126 [-0.134, -0.116]
profile-cosine rotation (locked)	384	0.1116	0.326	reference	
exhaustive 12-rotation DTW	384	0.1131	0.331	+0.0015 [+0.0012, +0.0019]	+0.005 [+0.003, +0.007]

Table 9: Registered A2: fused global + window Stage 1 vs global Stage 1, hub-corrected hybrid, paired (LOCAL-FULL; criterion met: yes)

K	coverage A1 $\rightarrow$ A2	MAP A1 $\rightarrow$ A2	Δ AP	Δ Hit@1 (points)
5	0.305 $\rightarrow$ 0.341	0.0847 $\rightarrow$ 0.1017	+0.0169 [+0.0151, +0.0188]	+3.1 [+2.5, +3.7]
10	0.380 $\rightarrow$ 0.419	0.0949 $\rightarrow$ 0.1126	+0.0176 [+0.0158, +0.0196]	+3.4 [+2.8, +3.9]
20	0.467 $\rightarrow$ 0.505	0.1105 $\rightarrow$ 0.1278	+0.0173 [+0.0154, +0.0192]	+3.1 [+2.6, +3.7]
30	0.522 $\rightarrow$ 0.560	0.1217 $\rightarrow$ 0.1388	+0.0171 [+0.0153, +0.0190]	+3.0 [+2.4, +3.5]
50	0.600 $\rightarrow$ 0.636	0.1374 $\rightarrow$ 0.1535	+0.0160 [+0.0143, +0.0179]	+2.6 [+2.1, +3.1]
100	0.706 $\rightarrow$ 0.740	0.1599 $\rightarrow$ 0.1744	+0.0145 [+0.0129, +0.0161]	+2.2 [+1.7, +2.7]
200	0.805 $\rightarrow$ 0.830	0.1824 $\rightarrow$ 0.1955	+0.0131 [+0.0116, +0.0147]	+2.3 [+1.8, +2.7]
500	0.908 $\rightarrow$ 0.917	0.2116 $\rightarrow$ 0.2217	+0.0101 [+0.0088, +0.0114]	+1.1 [+0.7, +1.6]

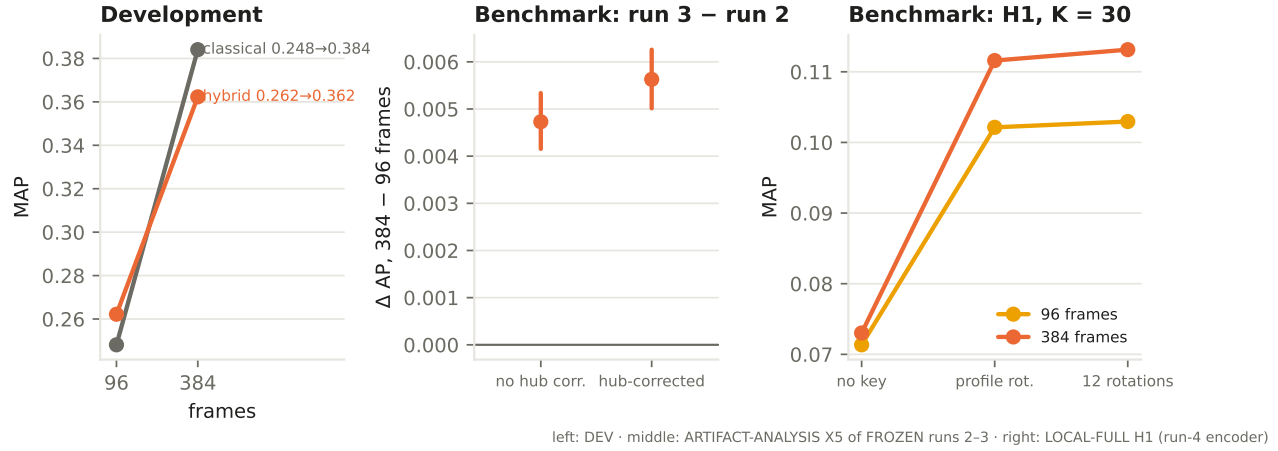


Figure 6: Alignment resolution and key handling across evidence tiers. Left: development (DEV). Middle: paired benchmark difference, run 3 minus run 2. Right: rerank-only benchmark MAP at $K = 30$ with the run-4 encoder (H1).

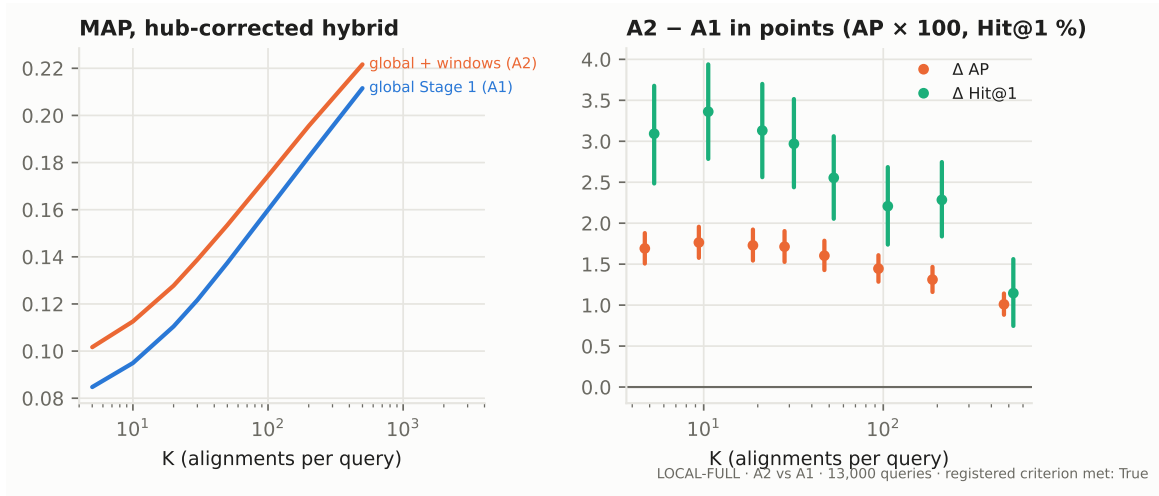


Figure 7: Registered A2: hub-corrected MAP with the global and the fused Stage 1 (left), and paired differences with 95% intervals (right).

- **Development pool** (120 candidates): tonal dispersion correlates with how often a candidate is a top-10 false positive, with Spearman -0.77 for classical alignment and -0.57 for the hybrid. The worst hub appears in 11 of 20 top-10 lists, against 1.7 expected by chance.
- **Probe reference.** The reference $\bar{a}(c)$ correlates with dispersion at -0.81 (96 frames) and -0.90 (384 frames). Subtracting it therefore targets exactly the static candidates.

Hubs at benchmark scale (P1). Figure 8 asks where the rank-1 non-covers of covered-but-misranked queries sit in the catalogue-wide distribution of $\bar{a}$.

- **Uncorrected:** the top decile of candidates supplies 28.4% of these false positives at $K = 30$ and 33.8% at $K = 500$, against 10% under uniformity.
- **Corrected:** the shares fall to 15.1% and 12.3%.
- **Why the value grows with K :** a larger shortlist admits more hubs, which is why the correction's value grows with K .

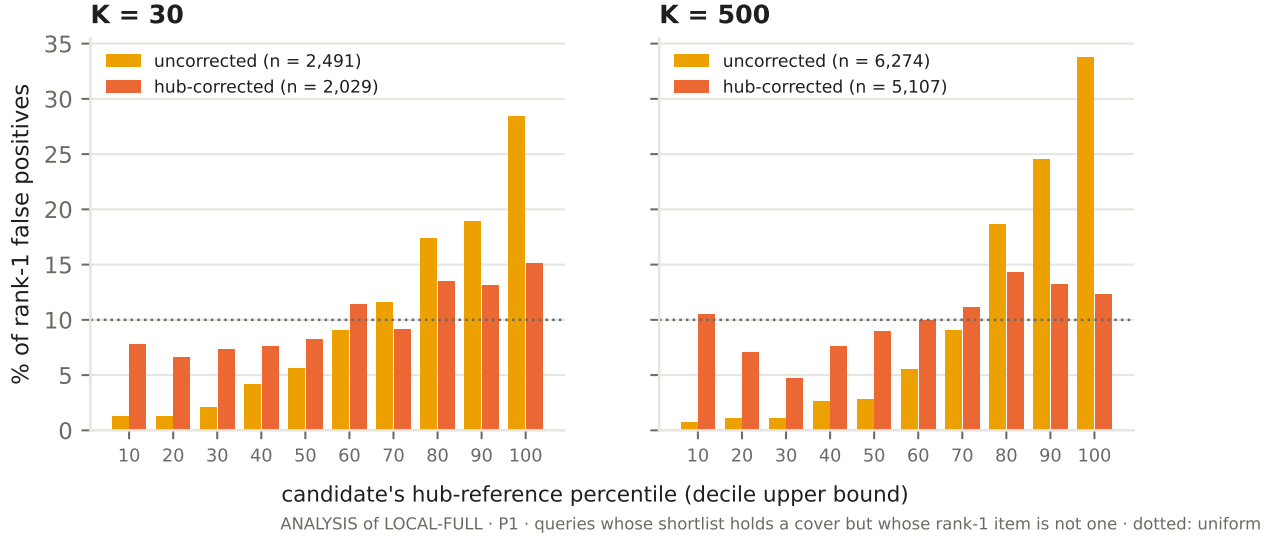


Figure 8: Where rank-1 false positives come from (P1): share by the candidate’s hub-reference decile, for queries whose shortlist contains a cover. Dotted line: 10% (uniform).

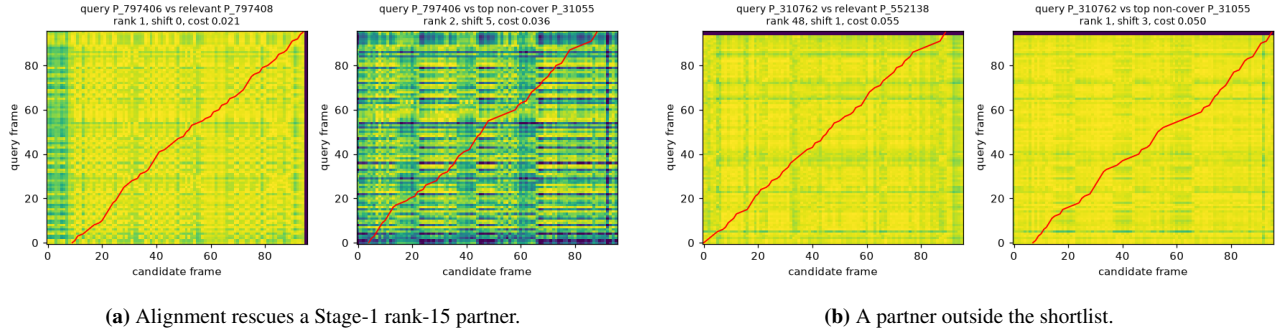


Figure 9: Development-protocol case studies: cost matrices with DTW paths (reports/figures).

- **The two stages have different hubs:** the Stage-1 k -occurrence of a candidate is nearly unrelated to its alignment reference (Spearman -0.17). Correcting the embedding stage was also not selected on calibration ($\lambda = 0$).

Case studies (DEV).

- **Alignment rescue** (Figure 9a): for query P_797406, Stage 1 ranks the true partner 15th. Alignment finds a clean diagonal path in the same key and moves it to rank 1.
- **Shortlist exclusion** (Figure 9b): query P_310762’s partner is ranked 48th by Stage 1. Exhaustive alignment would rank it 11th, but it never reaches the $K = 30$ shortlist.
- **Harmful reranking:** query P_207372’s partner falls from Stage-1 rank 5 to hybrid rank 22.

These cases illustrate the three classes. They are chosen examples, not estimates of frequency.

9 Accuracy–Computation Analysis

Table 10 gives the measured costs.

- **Stage 1:** embedding all 15,000 tracks takes 69 s and exact search for all queries 0.09 s.

Table 10: Measured runtime on the benchmark (194,987,000 query-candidate pairs); exhaustive alignment is alignment-only time; run 2 reused the run-1 alignment matrix

run	embed 15,000 tracks	exact search	K = 30 rerank	pairs aligned at K = 30	exhaustive alignment	hub probe reference
Run 1 (96 fr.)	69 s	0.09 s	69 s (5.3 ms/query)	390,000 (0.20%)	9,458 s (2.63 h)	-
Run 2 (96 fr.)	68 s	0.10 s	69 s (5.3 ms/query)	390,000 (0.20%)	9,458 s (2.63 h)	146 s
Run 3 (384 fr.)	68 s	0.07 s	494 s (38.0 ms/query)	390,000 (0.20%)	not run (about 42 h, estimated)	2220 s
Run 4 (384 fr.)	69 s	0.09 s	495 s (38.1 ms/query)	390,000 (0.20%)	not run (about 42 h, estimated)	2204 s

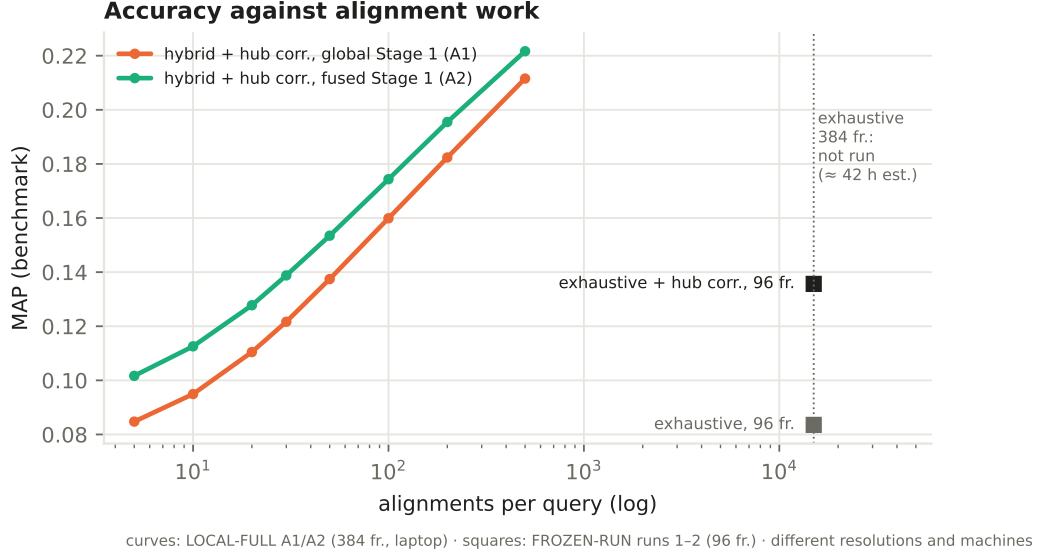


Figure 10: MAP against alignments per query. Curves: hybrid with global and fused Stage 1 (A1, A2; 384 frames). Squares: exhaustive alignment at 96 frames, with and without hub correction (runs 1–2). The 384-frame exhaustive system was not run.

- **Reranking at $K = 30$:** 69 s at 96 frames (5.3 ms per query) and 495 s at 384 frames (38 ms per query). It touches 390,000 of 1.95×10^8 pairs (0.2%).
- **Exhaustive alignment** at 96 frames took 9,458 s (2.63 h, alignment only). The hub-corrected variant adds a one-off probe reference of 146 s.
- **Not run:** exhaustive alignment at 384 frames, estimated at about 42 h.

Figure 10 places the systems by alignments per query. The hybrid curve crosses the 96-frame exhaustive system between $K = 50$ and $K = 100$ with the global Stage 1, and earlier with the fused one. A paired post-hoc comparison (Appendix, Table 12) gives, for the hub-corrected hybrid minus the exhaustive corrected system:

- $K = 50$: +0.0018 [-0.0046, +0.0081] AP (a tie);
- $K = 100$: +0.0243 [+0.0179, +0.0304], at 100 instead of 14,999 alignments per query.

Three caveats limit this comparison: it was defined after seeing A1; the two systems use different resolutions (384 vs. 96 frames); and the timings come from different machines. Neither faster nor more accurate settles the matter alone. The hybrid buys a large reduction in alignment work with a coverage risk that the decomposition makes explicit.

10 Development-vs-Benchmark Protocol Analysis

We asked the same paired question of both protocols four times (P2, Figure 11):

1. Hybrid minus exhaustive alignment, run-1 encoder: DEV +0.014 [-0.009, +0.042]; benchmark -0.065 [-0.071, -0.059].

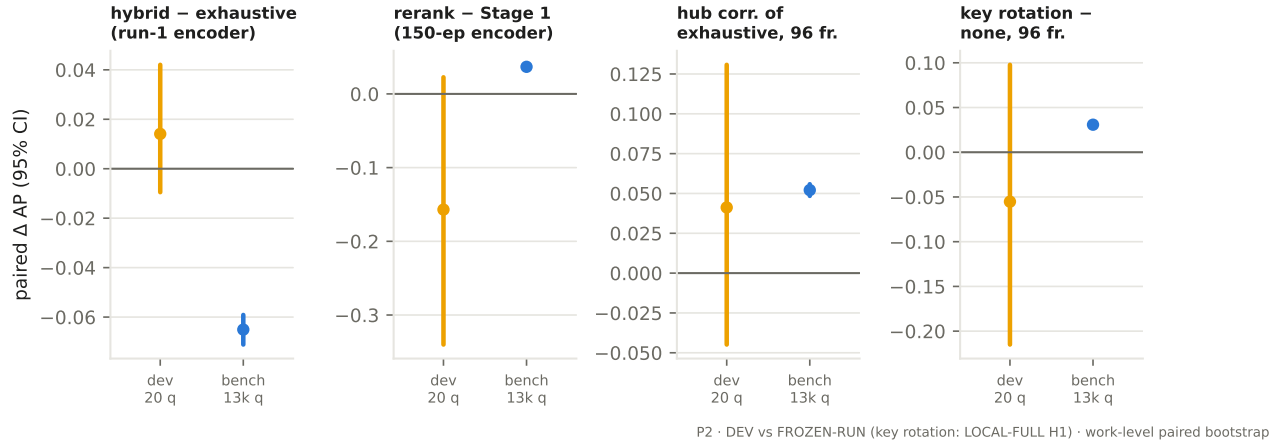


Figure 11: The same paired comparison on the development protocol (20 queries) and on the benchmark (13,000 queries). Bars: 95% work-level intervals.

2. Reranking minus Stage 1, 150-epoch encoder: DEV -0.157 [-0.340, +0.023]; benchmark +0.037 [+0.034, +0.039].
3. Hub correction of exhaustive alignment: DEV +0.041 [-0.045, +0.131]; benchmark +0.052 [+0.048, +0.056].
4. Key rotation minus none: DEV -0.055 [-0.215, +0.098]; benchmark +0.031 [+0.028, +0.034].

The development interval included zero in 4 of 4 cases, and its point estimate had the wrong sign in 3. The benchmark interval excluded zero in 4 of 4. The development protocol did not “lie”. It was inconclusive, and its point estimates were read as conclusions.

Why the protocols diverge. Three structural differences explain it.

1. **The shortlist covers different fractions of the pool.** $K = 30$ is 25% of a 119-item development pool but 0.2% of the benchmark’s 14,999. Development shortlist recall was 0.75–0.90, against 0.021–0.136 on the benchmark (X6). The development protocol therefore measures the reranker given near-perfect coverage, a regime the benchmark never enters at $K = 30$.
2. **One relevant item against twelve.** With one relevant item, AP equals reciprocal rank, so the metric rewards only the first cover. With twelve, MAP also rewards recovering the rest, which reranking a short list cannot do.
3. **Sample size.** 20 query works give intervals of ± 0.05 –0.2 AP. The effects that matter at scale are 0.005–0.05.

A useful development protocol for two-stage CSI must reproduce the benchmark’s candidate-to-relevant ratio at the short-list, and must be large enough for its intervals to exclude effects of that size.

11 Discussion

What we show.

- At $K = 30$ the coarse stage is the main limitation, and improving it produced most of the observed gains.
- The dominant failure moves from exclusion to misranking as K grows, while accuracy keeps rising to $K = 500$.
- Hub correction transfers from calibration to 13,000 queries and matters more at large K , because hubs concentrate among high-reference candidates.
- Fusing global and window embeddings improves the whole system. Per-query adaptive K does not.
- 384-frame alignment helps on the benchmark at $K = 30$, by much less than on development.
- A 20-query development protocol is inconclusive on every comparison we tested.

What we suspect.

- The location of the crossover ($K \approx 100\text{--}200$) depends on the encoder and the reranker. A stronger Stage 1 should move it to smaller K .
- Adaptive K likely fails because the rank of the first cover is heavy-tailed even for queries predicted to be hard, so extra alignments are better spent on queries that are already covered.
- Exhaustive alignment at 384 frames would probably widen its lead over the $K = 30$ hybrid, as development suggests. It might not beat the hybrid at larger K .

What we did not test.

- Exhaustive 384-frame alignment on the benchmark (not run; about 42 h estimated).
- Any other dataset or feature type.
- A strong external encoder. Da-TACOS has no audio, so audio models such as Discogs-VINet [Araz et al., 2024] cannot be run on it. Pitch-class models such as MOVE and Re-MOVE [Yesiler et al., 2020a,b] can, because Da-TACOS distributes their CREMA-PCP input.
- Fusion settings other than the one registered.
- Local (Qmax-style) alignment.

12 Limitations

- **Absolute accuracy.** Qmax reaches 0.333 MAP on the same HPCP input [Yesiler et al., 2019]; our best locked system reaches 0.136, and the best measured configuration 0.222. Findings about a weak system may not transfer to strong ones.
- **Scope.** HPCP only; one dataset; features without audio; training works with two recordings each; a single global vector in the frozen Stage 1; whole-query rather than local alignment.
- **The development protocol** is not representative of the benchmark (Section 10).
- **Provenance.** The frozen runs record commits with uncommitted changes (“dirty”). The local-full runs were made on a different machine, also from working trees with uncommitted changes, which is recorded in each run’s environment file. A1 reproduces run 4 exactly, and cross-run equality checks pass, which bounds but does not remove this concern.
- **Registration timing.** A2 was registered after its coverage effect (F1) was known, though before any end-to-end result. Only one fusion setting was evaluated.
- **Reproduction boundaries.** Checkpoints are not redistributed, since derived weights inherit the non-commercial licence. They are reproducible bit for bit from the committed configuration.

13 Future Work

In order of what the evidence asks for:

1. Re-run the decomposition with a strong open encoder that accepts Da-TACOS features, such as MOVE or Re-MOVE on CREMA-PCP [Yesiler et al., 2020a,b], to test whether the crossover and hub findings generalise.
2. Choose fusion settings (window count, length, weight) on calibration works, and evaluate test-time key rotation end to end.
3. Improve discrimination inside large shortlists, where most failures are at $K \geq 200$: stronger hub-aware scoring, local alignment.

4. Run exhaustive 384-frame alignment once, to close the gap in Figure 10.
5. Build a development protocol whose candidate-to-relevant ratio matches the benchmark’s.

These are proposals. None has been run unless it is reported above.

14 Conclusion

In coarse-to-fine cover retrieval, candidate selection is a measurable bottleneck, not an implementation detail. On DaTACOS, a transparent two-stage system is limited mainly by coverage at $K = 30$, and improving the coarse stage produced most of its gains. Enlarging the shortlist keeps paying but hands the bottleneck to the reranker. Hub correction and a fused global–window first stage are among the few changes that help at every K . The same analysis on a 20-query development set would have supported none of these conclusions.

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Table 11: Coverage C(K): % of queries with ≥ 1 cover in the top K (work-level 95% CI)

ranking	K = 10	K = 30	K = 100	K = 500	K = 1000
Stage 1 · 1,500 works, 60 ep	9.6 [8.9, 10.4]	18.7 [17.5, 19.8]	37.1 [35.7, 38.5]	73.0 [72.0, 74.0]	86.5 [85.8, 87.2]
Stage 1 · 4,780 works, 60 ep	24.3 [22.9, 25.7]	39.0 [37.4, 40.5]	59.4 [58.0, 60.8]	86.0 [85.2, 86.8]	93.4 [92.9, 93.9]
Stage 1 · 4,780 works, 150 ep	38.0 [36.3, 39.6]	52.2 [50.5, 53.8]	70.6 [69.4, 71.9]	90.8 [90.2, 91.4]	95.4 [95.0, 95.8]
exhaustive alignment, 96 fr.	35.2 [33.5, 36.9]	42.4 [40.7, 44.1]	52.4 [50.7, 54.1]	73.2 [71.8, 74.5]	83.6 [82.6, 84.6]
exhaustive alignment + hub corr.	48.1 [46.5, 49.6]	55.8 [54.3, 57.2]	66.1 [64.8, 67.5]	82.9 [82.0, 83.8]	90.2 [89.6, 90.9]

Table 12: POST-HOC, not registered: hub-corrected hybrid at K vs exhaustive corrected alignment (run 2 classical_alignment_hubcorr (96 frames, lambda 0.5)); paired Δ , work-level 95% CI

K	alignments per query	Δ AP	Δ Hit@1
5	5 vs 14,999	-0.0509 [-0.0576, -0.0443]	-0.091 [-0.104, -0.078]
10	10 vs 14,999	-0.0407 [-0.0473, -0.0342]	-0.051 [-0.064, -0.038]
20	20 vs 14,999	-0.0252 [-0.0316, -0.0188]	-0.010 [-0.022, +0.002]
30	30 vs 14,999	-0.0140 [-0.0204, -0.0075]	+0.016 [+0.003, +0.027]
50	50 vs 14,999	+0.0018 [-0.0046, +0.0081]	+0.048 [+0.036, +0.059]
100	100 vs 14,999	+0.0243 [+0.0179, +0.0304]	+0.087 [+0.076, +0.098]
200	200 vs 14,999	+0.0467 [+0.0405, +0.0528]	+0.124 [+0.113, +0.134]
500	500 vs 14,999	+0.0759 [+0.0701, +0.0817]	+0.165 [+0.155, +0.175]

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A Supplementary tables and figures

Table 13: Adaptive vs fixed K at equal budget (ANALYSIS of LOCAL-FULL)

menu	budget K	realised mean K	policy MAP	fixed MAP	Δ AP	Δ Hit@1
(10, 30, 100)	10	10.0	0.0949	0.0949	+0.0000 [+0.0000, +0.0000]	+0.000 [+0.000, +0.000]
(10, 30, 100)	20	20.1	0.1168	0.1105	+0.0064 [+0.0054, +0.0073]	-0.004 [-0.008, -0.000]
(10, 30, 100)	30	29.9	0.1259	0.1217	+0.0043 [+0.0027, +0.0059]	-0.027 [-0.032, -0.023]
(10, 30, 100)	50	50.1	0.1433	0.1374	+0.0058 [+0.0041, +0.0076]	-0.036 [-0.041, -0.030]
(10, 30, 100)	100	100.0	0.1599	0.1599	-0.0000 [-0.0000, +0.0000]	+0.000 [+0.000, +0.000]
(10, 50, 200)	10	10.0	0.0949	0.0949	+0.0000 [-0.0000, +0.0000]	+0.000 [+0.000, +0.000]
(10, 50, 200)	20	20.0	0.1168	0.1105	+0.0063 [+0.0047, +0.0079]	-0.021 [-0.026, -0.017]
(10, 50, 200)	30	30.1	0.1289	0.1217	+0.0072 [+0.0059, +0.0086]	-0.016 [-0.021, -0.012]
(10, 50, 200)	50	49.7	0.1377	0.1374	+0.0003 [-0.0005, +0.0010]	-0.014 [-0.017, -0.011]
(10, 50, 200)	100	100.2	0.1600	0.1599	+0.0001 [-0.0016, +0.0018]	-0.044 [-0.050, -0.039]

Table 14: Predicting ‘no cover in the top 30’ from label-free Stage-1 signals: AUROC (base rate 47.8%; below 0.5 = higher for coverable queries; ANALYSIS of LOCAL-FULL)

signal	AUROC
cross-fitted logistic model (all signals)	0.760 [0.748, 0.771]
entropy top100	0.729
z top1	0.281
std top10	0.303
mean top100	0.696
margin 1 10	0.304
mean top10	0.655
row std	0.620
top1	0.601
margin 1 2	0.403
top1 kocc10	0.563
query kocc10	0.523
top1 hub reference	0.481

Table 15: Stage-1 variants: coverage at K = 30 and 100, paired Δ vs the reference, points (ANALYSIS of LOCAL-FULL)

Stage 1	coverage@30 (%)	coverage@100 (%)	Δ @30	Δ @100
global, 150 epochs, 4,780 works (reference)	52.2 [50.5, 53.8]	70.6 [69.4, 71.9]		
global + 3 windows, fused	56.0 [54.4, 57.6]	74.0 [72.8, 75.2]	+3.8 [+3.2, +4.5]	+3.3 [+2.7, +4.0]
test-time 12 key rotations	53.6 [51.9, 55.1]	71.7 [70.4, 72.9]	+1.4 [+0.8, +2.0]	+1.0 [+0.5, +1.6]
Stage-1 hub corr. (calibration chose $\lambda = 0$: identical to reference)	52.2 [50.5, 53.8]	70.6 [69.4, 71.9]	+0.0 [+0.0, +0.0]	+0.0 [+0.0, +0.0]
3 windows only (MaxSim)	50.7 [49.0, 52.3]	69.2 [67.8, 70.5]	-1.5 [-2.5, -0.5]	-1.4 [-2.3, -0.5]
encoder: 60 epochs	39.0 [37.4, 40.5]	59.4 [58.0, 60.8]	-13.2 [-14.4, -12.1]	-11.2 [-12.2, -10.2]
encoder: 1,500 training works	18.7 [17.5, 19.8]	37.1 [35.7, 38.5]	-33.5 [-35.0, -32.1]	-33.5 [-34.8, -32.2]

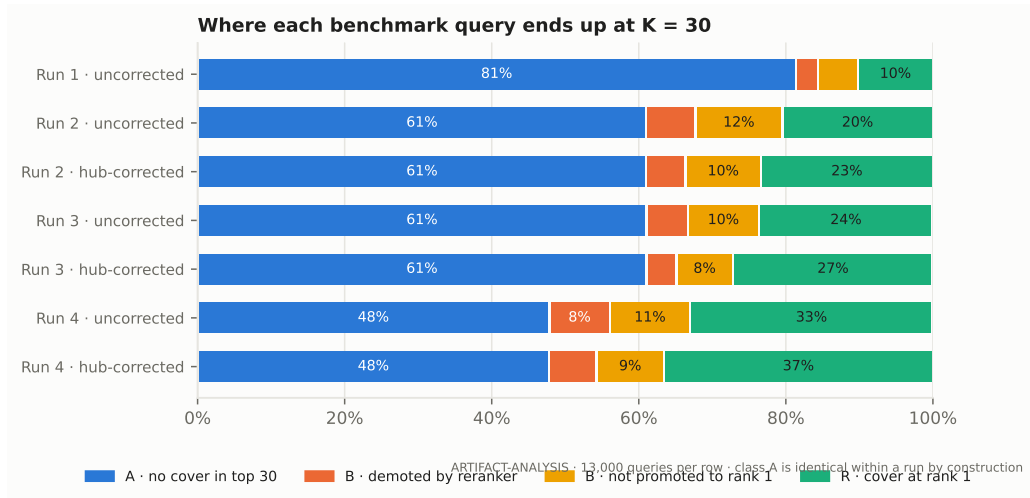


Figure 12: Failure classes at $K = 30$ for the four runs (X2).

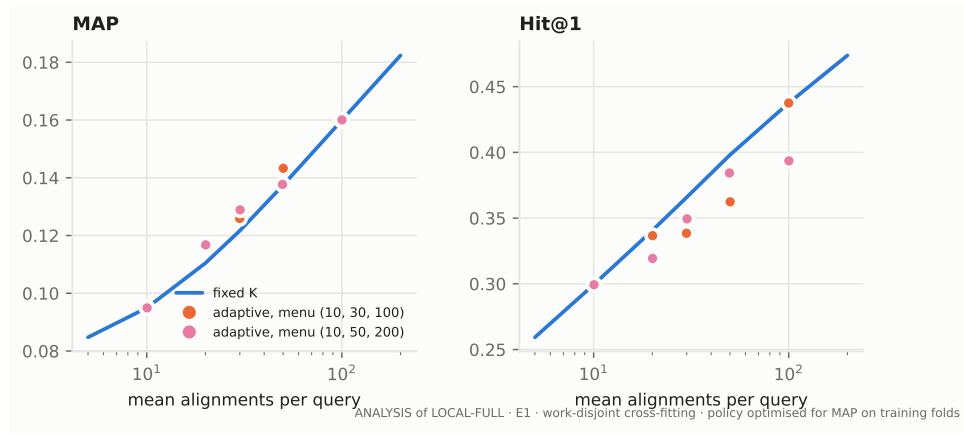


Figure 13: Adaptive against fixed K at equal mean cost (E1, cross-fitted).

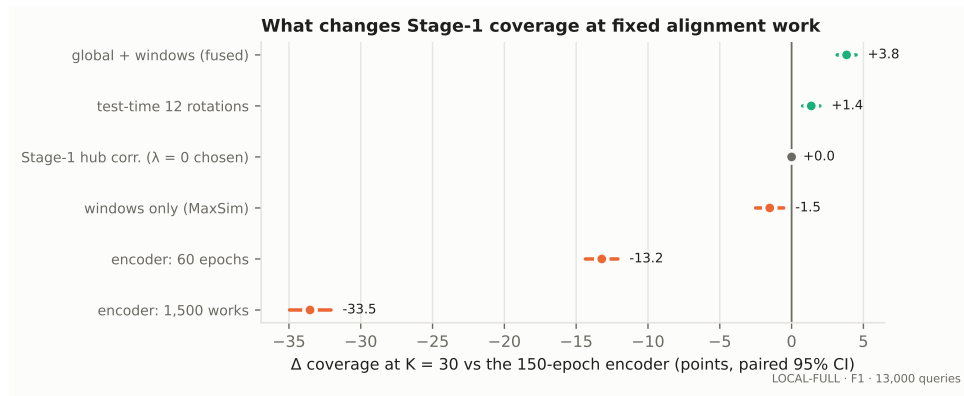


Figure 14: Paired change in coverage at $K = 30$ for Stage-1 variants (F1).